535/2
PHYSICS
Paper 2
Jul. /Aug. 2026
2 hours

MASAKA DIOCESE EXAMINATIONS BOARD

UCE Joint Mock Examinations 2026

PHYSICS

Paper 2

(Practical)

2 hours

INSTRUCTIONS TO CANDIADATES:

This paper consists **two** *examination items.*

Respond to **one** *item only.*

Any additional items answered will **not** *be scored.*

You are **not** *allowed to start working with the apparatus for the first* **15 minute**. *This time is to enable you read the items thoroughly, check whether you have all the apparatus needed for the investigation and plan appropriately.*

Responses to the item **must** *be written in the answer booklet(s) provided.*

Indicate the item number you are responding to, clearly at the top of the page.

A graph paper will be provided.

Mathematical tables and silent non – programmable calculators may be used.

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Item 1

A factory that manufactures padlocks has received numerous complaints from customers that the keys break easily during use. The company's management suspects that the material used to manufacture the keys may have an unsuitable density and is considering changing the supplier.

For the production of durable and long-lasting keys, the material used should have a density between 9.0 g cm^{-3} and 12.0 g cm^{-3}, inclusive.

The material used to manufacture the keys has the same density as that of the pendulum bob provided.

Task

Design and carry out an investigation to advise the company's management on whether they should retain the current supplier or source the material from a new supplier.

Hint

In your report, include; the aim, variables (independent, dependent and controlled), hypothesis, procedure, possible sources of errors, precautions to the stated errors, data presentation (at least 5 data sets), data analysis (using graph), and advise/ recommendation.

Item 2

In an effort to conserve the environment, a household has been advised to use a solar concentrator for cooking. The appropriate solar concentrator should have a focal length between 135 cm and 165 cm inclusive.

A trader is offering a solar concentrator at a discounted price, but the household is uncertain whether it is suitable for their needs. The trader states that the focal length of the solar concentrator is 15 times the focal length of a standard concave mirror.

You have been provided with the standard concave mirror.

Task

Design and carry out an investigation to advice the household on whether they should purchase the solar concentrator.

Hint

In your report, include; the aim, variables (independent, dependent and controlled), hypothesis, procedure, possible sources of errors, precautions to the stated errors, data presentation (at least 5 data sets), data analysis (using graph), and advise/ recommendation.

END

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